

KADI SARVA VISHWAVIDYALAYA
B. E. Semester – VII (ATKT) Examination (March-2024)
Distributed System (CT704A-N)

Branch: CE/ IT/ CSE
Time: 12:00 am to 3:00 pm
Instructions:

Date: 30-03-2024
Total Marks: 70

1. Answer each section in separate Answer sheet.
2. All questions are compulsory.
3. Indicate clearly, the options you attempt along with its respective question number.
4. Use the last page of main supplementary for rough work.

Section – I

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|----------|---|---|
| Q 1: (A) | Describe group communication. | 5 |
| Q 1: (B) | What is IPC? Briefly explains message passing and shared memory System. | 5 |
| Q 1: (C) | Briefly explain distributed system computing models. | 5 |

OR

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|----------|---|---|
| Q 1: (C) | Compare Workstation Server Model v/s Processor Pool Model | 5 |
| Q 2: (A) | Discuss about ATM reference model with its functionality. | 5 |
| Q 2: (B) | Explain different message buffering strategies. | 5 |

OR

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|----------|---|---|
| Q 2: (A) | Explain failure handling mechanism of IPC? | 5 |
| Q 2: (B) | Difference between FLIP and VMTP | 5 |
| Q 3: (A) | What is RPC? RPC basic operation with diagrams | 5 |
| Q 3: (B) | Give Overview of Security techniques and explain digital signature. | 5 |

OR

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| Q 3: (A) | Explain RPC communication protocols with necessary figure. | 5 |
| Q 3: (B) | Explain Bully election and ring election algorithm. | 5 |

Section – II

- Q 4: (A) List and explain conditions which are necessary and sufficient for a deadlock to occur in a system 5
- Q 4: (B) Explain centralized algorithm for implementing critical section and Mutual exclusion. 5
- Q 4: (C) What are various issues in designing load sharing algorithm. 5

OR

- Q 4: (C) What are the various issues while designing load balancing algorithm? Explain any one in detail. 5
- Q 5: (A) What is process migration? List steps for it. What is advantage of process migration? 5
- Q 5: (B) Draw and explain DSM architecture. 5

OR

- Q 5: (A) Discuss various issues in designing the DSM system. 5
- Q 5: (B) Write and explain algorithm of constructing a DFS spanning tree with a specified root 5
- Q 6: (A) Explain DFS file models. 5
- Q 6: (B) What are the features of file sharing semantics? 5

OR

- Q 6: (A) Compare file transfer models. 5
- Q 6: (B) Explain SOA in detail 5